Example 3-14: Verification of Stokes's Theorem

For vector field $\mathbf{B} = \hat{\mathbf{z}}\cos\phi/r$, verify Stokes's theorem for a segment of a cylindrical surface defined by $r = 2$, $\pi/3 \le \phi \le \pi/2$, and $0 \le z \le 3$ (**Fig. 3-25**).

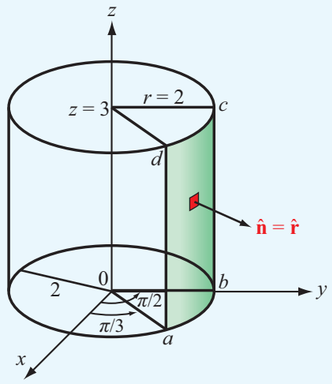


Figure 3-25 Geometry of Example 3-14.

Solution: Stokes's theorem states that

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} = \oint_C \mathbf{B} \cdot d\mathbf{l}.$$

Left-hand side: With $\mathbf{B}$ having only a component $B_z = \cos\phi/r$, use of the expression for $\nabla \times \mathbf{B}$ in cylindrical coordinates in Appendix C gives

$$\begin{aligned}\nabla \times \mathbf{B} &= \hat{\mathbf{r}}\left(\frac{1}{r}\frac{\partial B_z}{\partial \phi} - \frac{\partial B_\phi}{\partial z}\right) + \hat{\boldsymbol{\phi}}\left(\frac{\partial B_r}{\partial z} - \frac{\partial B_z}{\partial r}\right) \\ &\quad + \hat{\mathbf{z}}\frac{1}{r}\left(\frac{\partial}{\partial r}(rB_\phi) - \frac{\partial B_r}{\partial \phi}\right) \\ &= \hat{\mathbf{r}}\frac{1}{r}\frac{\partial}{\partial \phi}\left(\frac{\cos\phi}{r}\right) - \hat{\boldsymbol{\phi}}\frac{\partial}{\partial r}\left(\frac{\cos\phi}{r}\right) \\ &= -\hat{\mathbf{r}}\frac{\sin\phi}{r^2} + \hat{\boldsymbol{\phi}}\frac{\cos\phi}{r^2}.\end{aligned}$$

The integral of $\nabla \times \mathbf{B}$ over the specified surface S is

$$\begin{aligned}\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} &= \int_{z=0}^{3}\int_{\phi=\pi/3}^{\pi/2}\left(-\hat{\mathbf{r}}\frac{\sin\phi}{r^2} + \hat{\boldsymbol{\phi}}\frac{\cos\phi}{r^2}\right)\cdot \hat{\mathbf{r}} r\, d\phi\, dz \\ &= \int_0^3\int_{\pi/3}^{\pi/2} -\frac{\sin\phi}{r}\, d\phi\, dz = -\frac{3}{2r} = -\frac{3}{4}.\end{aligned}$$

Right-hand side: The surface S is bounded by contour $C = abcd$ shown in **Fig. 3-25**. The direction of C is chosen so that it is compatible with the surface normal $\hat{\mathbf{r}}$ by the right-hand rule. Hence,

$$\begin{aligned}\oint_C \mathbf{B} \cdot d\mathbf{l} &= \int_a^b \mathbf{B}_{ab} \cdot d\mathbf{l} + \int_b^c \mathbf{B}_{bc} \cdot d\mathbf{l} \\ &\quad + \int_c^d \mathbf{B}_{cd} \cdot d\mathbf{l} + \int_d^a \mathbf{B}_{da} \cdot d\mathbf{l},\end{aligned}$$

where $\mathbf{B}_{ab}$, $\mathbf{B}_{bc}$, $\mathbf{B}_{cd}$, and $\mathbf{B}_{da}$ are the field $\mathbf{B}$ along segments ab, bc, cd, and da, respectively. Over segment ab, the dot product of $\mathbf{B}_{ab} = \hat{\mathbf{z}}(\cos\phi)/2$ and $d\mathbf{l} = \hat{\boldsymbol{\phi}} r\, d\phi$ is zero, and the same is true for segment cd. Over segment bc, $\phi = \pi/2$; hence, $\mathbf{B}_{bc} = \hat{\mathbf{z}}(\cos\pi/2)/2 = 0$. For the last segment, $\mathbf{B}_{da} = \hat{\mathbf{z}}(\cos\pi/3)/2 = \hat{\mathbf{z}}/4$ and $d\mathbf{l} = \hat{\mathbf{z}}\, dz$. Hence,

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \int_d^a \left(\hat{\mathbf{z}}\frac{1}{4}\right)\cdot \hat{\mathbf{z}}\, dz = \int_3^0 \frac{1}{4}\, dz = -\frac{3}{4},$$

which is the same as the result obtained by evaluating the left-hand side of Stokes's equation.

Exercise 3-18: Find $\nabla \times \mathbf{A}$ at $(2, 0, 3)$ in cylindrical coordinates for the vector field

$$\mathbf{A} = \hat{\mathbf{r}}10e^{-2r}\cos\phi + \hat{\mathbf{z}}10\sin\phi.$$

Answer: (See EM.)

$$\nabla \times \mathbf{A} = \left(\hat{\mathbf{r}}\frac{10\cos\phi}{r} + \frac{\hat{\mathbf{z}}10e^{-2r}}{r}\sin\phi\right)\Big|_{(2,0,3)} = \hat{\mathbf{r}}5.$$

Exercise 3-19: Find $\nabla \times \mathbf{A}$ at $(3, \pi/6, 0)$ in spherical coordinates for the vector field $\mathbf{A} = \hat{\boldsymbol{\theta}}12\sin\theta$.

Answer: (See EM.)

$$\nabla \times \mathbf{A} = \hat{\boldsymbol{\phi}}\frac{12\sin\theta}{R}\Big|_{(3,\pi/6,0)} = \hat{\boldsymbol{\phi}}2.$$

例 3-14：斯托克斯定理的验证

对于矢量场 B = ẑ cos ϕ/r，验证由 r = 2、π/3 ≤ ϕ ≤ π/2 和 0 ≤ z ≤ 3 定义的一段圆柱表面的斯托克斯定理（图 3-25）。

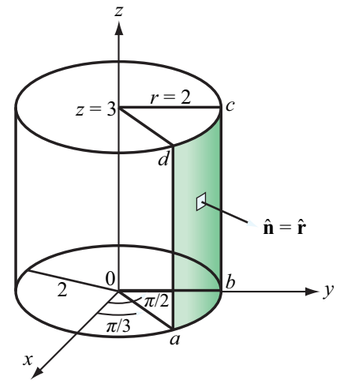


图 3-25 示例 3-14 的几何形状。

解：斯托克斯定理指出

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} = \oint_C \mathbf{B} \cdot d\mathbf{l}.$$

左侧：当 B 仅具有分量 $B_z = \cos\phi/r$ 时，使用附录 C 中的柱坐标中的 ∇ × B 表达式可得出

$$\begin{aligned}\nabla \times \mathbf{B} &= \hat{\mathbf{r}}\left(\frac{1}{r}\frac{\partial B_z}{\partial \phi} - \frac{\partial B_\phi}{\partial z}\right) + \hat{\boldsymbol{\phi}}\left(\frac{\partial B_r}{\partial z} - \frac{\partial B_z}{\partial r}\right) \\ &\quad + \hat{\mathbf{z}}\frac{1}{r}\left(\frac{\partial}{\partial r}(rB_\phi) - \frac{\partial B_r}{\partial \phi}\right) \\ &= \hat{\mathbf{r}}\frac{1}{r}\frac{\partial}{\partial \phi}\left(\frac{\cos\phi}{r}\right) - \hat{\boldsymbol{\phi}}\frac{\partial}{\partial r}\left(\frac{\cos\phi}{r}\right) \\ &= -\hat{\mathbf{r}}\frac{\sin\phi}{r^2} + \hat{\boldsymbol{\phi}}\frac{\cos\phi}{r^2}.\end{aligned}$$

∇ × B 在指定表面 S 上的积分为

$$\begin{aligned}\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} &= \int_{z=0}^{3}\int_{\phi=\pi/3}^{\pi/2}\left(-\hat{\mathbf{r}}\frac{\sin\phi}{r^2} + \hat{\boldsymbol{\phi}}\frac{\cos\phi}{r^2}\right)\cdot \hat{\mathbf{r}} r\, d\phi\, dz \\ &= \int_0^3\int_{\pi/3}^{\pi/2} -\frac{\sin\phi}{r}\, d\phi\, dz = -\frac{3}{2r} = -\frac{3}{4}.\end{aligned}$$

右侧：曲面 S 以轮廓 C = abcd 为界，如图 3-25 所示。选择 C 的方向，使其与根据右手定则的表面法线兼容。因此，

$$\begin{aligned}\oint_C \mathbf{B} \cdot d\mathbf{l} &= \int_a^b \mathbf{B}_{ab} \cdot d\mathbf{l} + \int_b^c \mathbf{B}_{bc} \cdot d\mathbf{l} \\ &\quad + \int_c^d \mathbf{B}_{cd} \cdot d\mathbf{l} + \int_d^a \mathbf{B}_{da} \cdot d\mathbf{l},\end{aligned}$$

其中 B_{ab}、B_{bc}、B_{cd} 和 B_{da} 分别是沿线段 ab、bc、cd 和 da 的域 B。在线段 ab 上，$B_{ab} = \hat{\mathbf{z}}(\cos\phi)/2$ 和 $d\mathbf{l} = \hat{\boldsymbol{\phi}} r\, d\phi$ 的点积为零，线段 cd 也是如此。过BC段，$\phi = \pi/2$；因此，$B_{bc} = \hat{\mathbf{z}}(\cos\pi/2)/2 = 0$。对于最后一段，$B_{da} = \hat{\mathbf{z}}(\cos\pi/3)/2 = \hat{\mathbf{z}}/4$ 和 $d\mathbf{l} = \hat{\mathbf{z}}\, dz$。因此，

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \int_d^a \left(\hat{\mathbf{z}}\frac{1}{4}\right)\cdot \hat{\mathbf{z}}\, dz = \int_3^0 \frac{1}{4}\, dz = -\frac{3}{4},$$

这与评估斯托克斯方程左侧得到的结果相同。

练习 3-18：在向量场的柱坐标中求 (2, 0, 3) 处的 ∇ × A

$$\mathbf{A} = \hat{\mathbf{r}}10e^{-2r}\cos\phi + \hat{\mathbf{z}}10\sin\phi.$$

答案：（参见EM。）

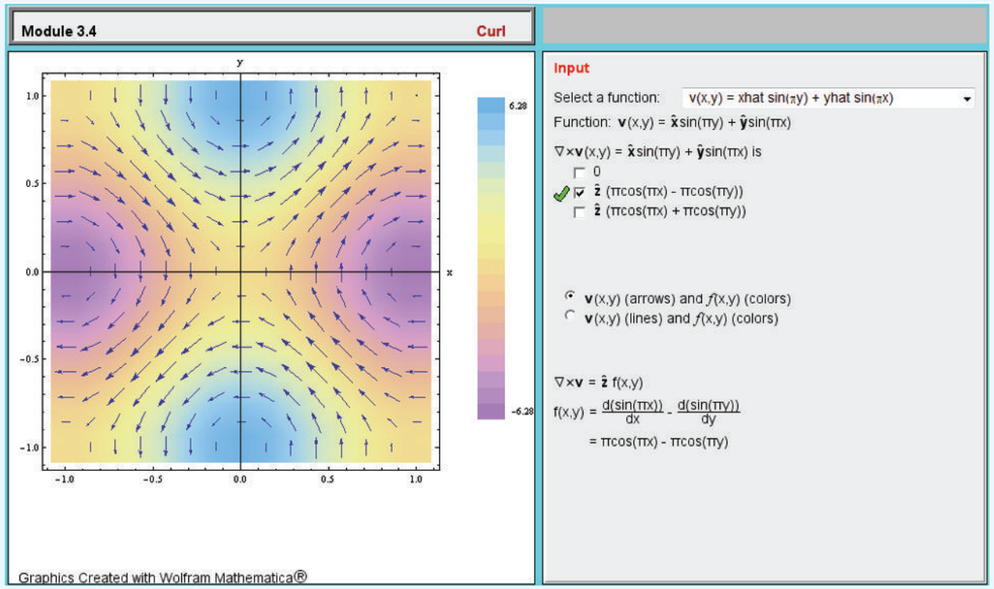
$$\nabla \times \mathbf{A} = \left(\hat{\mathbf{r}}\frac{10\cos\phi}{r} + \frac{\hat{\mathbf{z}}10e^{-2r}}{r}\sin\phi\right)\Big|_{(2,0,3)} = \hat{\mathbf{r}}5.$$

练习 3-19：在球坐标系中针对矢量场 $\mathbf{A} = \hat{\boldsymbol{\theta}}12\sin\theta$ 求 $(3, \pi/6, 0)$ 处的 ∇ × A。

Answer: (See EM.)

$$\nabla \times \mathbf{A} = \hat{\boldsymbol{\phi}}\frac{12\sin\theta}{R}\Big|_{(3,\pi/6,0)} = \hat{\boldsymbol{\phi}}2.$$

Module 3.4 Curl Select a vector $\mathbf{f}(x,y)$, evaluate its curl, and display both in the x–y plane.



3-7 Laplacian Operator

In later chapters, we sometimes deal with problems involving multiple combinations of operations on scalars and vectors. A frequently encountered combination is the divergence of the gradient of a scalar. For a scalar function V defined in Cartesian coordinates, its gradient is

$$\nabla V = \hat{\mathbf{x}}\frac{\partial V}{\partial x} + \hat{\mathbf{y}}\frac{\partial V}{\partial y} + \hat{\mathbf{z}}\frac{\partial V}{\partial z} = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z = \mathbf{A}, \tag{3.108}$$

where we defined a vector $\mathbf{A}$ with components $A_x = \partial V/\partial x$, $A_y = \partial V/\partial y$, and $A_z = \partial V/\partial z$. The divergence of ∇V is

$$\nabla\cdot(\nabla V) = \nabla\cdot\mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}. \tag{3.109}$$

For convenience, $\nabla\cdot(\nabla V)$ is called the ***Laplacian*** of V and is denoted by $\nabla^2 V$ (the symbol ∇^2 is pronounced "del square").

That is,

$$\nabla^2 V = \nabla\cdot(\nabla V) = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}. \tag{3.110}$$

As we can see from Eq. (3.110), the Laplacian of a scalar function is a scalar. Expressions for $\nabla^2 V$ in cylindrical and spherical coordinates are given in Appendix C.

The Laplacian of a scalar can be used to define the Laplacian of a vector. For a vector $\mathbf{E}$ specified in Cartesian coordinates as

$$\mathbf{E} = \hat{\mathbf{x}}E_x + \hat{\mathbf{y}}E_y + \hat{\mathbf{z}}E_z, \tag{3.111}$$

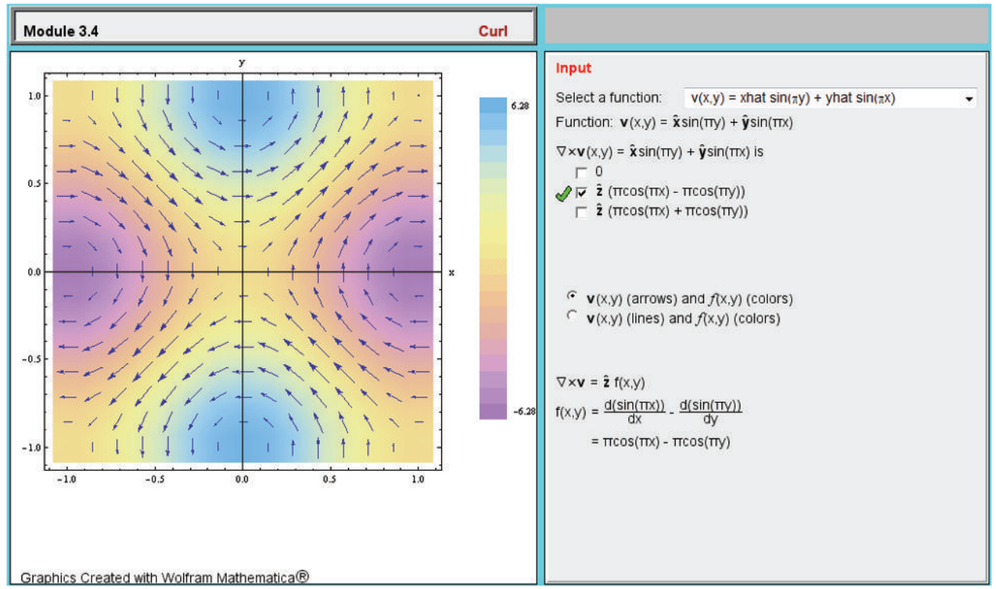
the Laplacian of $\mathbf{E}$ is

$$\nabla^2\mathbf{E} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\mathbf{E} = \hat{\mathbf{x}}\nabla^2 E_x + \hat{\mathbf{y}}\nabla^2 E_y + \hat{\mathbf{z}}\nabla^2 E_z. \tag{3.112}$$

Thus, in Cartesian coordinates the Laplacian of a vector is a vector whose components are equal to the Laplacians of the vector components. Through direct substitution, it can be

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模块 3.4 旋度 选择一个向量 f(x, y)，评估其旋度，并将两者显示在 x–y 平面中。



3-7 拉普拉斯算子

在后面的章节中，我们有时会处理涉及标量和向量运算的多种组合的问题。经常遇到的组合是标量梯度的散度。对于在笛卡尔坐标系中定义的标量函数 V，其梯度为

$$\nabla V = \hat{\mathbf{x}}\frac{\partial V}{\partial x} + \hat{\mathbf{y}}\frac{\partial V}{\partial y} + \hat{\mathbf{z}}\frac{\partial V}{\partial z} = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z = \mathbf{A}, \tag{3.108}$$

其中我们定义了一个向量 A，其分量为 $A_x = \partial V/\partial x$、$A_y = \partial V/\partial y$ 和 $A_z = \partial V/\partial z$。 ∇V 的散度为

$$\nabla\cdot(\nabla V) = \nabla\cdot\mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}. \tag{3.109}$$

为了方便起见，$\nabla\cdot(\nabla V)$被称为V的拉普拉斯算子，记为 $\nabla^2 V$（符号∇^2发音为“del square”）。

那是，

$$\nabla^2 V = \nabla\cdot(\nabla V) = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}. \tag{3.110}$$

正如我们从等式中可以看到的那样。（3.110），标量函数的拉普拉斯算子是标量。附录 C 中给出了柱坐标和球坐标中 $\nabla^2 V$ 的表达式。

标量的拉普拉斯算子可用于定义向量的拉普拉斯算子。对于笛卡尔坐标中指定的向量 E

$$\mathbf{E} = \hat{\mathbf{x}}E_x + \hat{\mathbf{y}}E_y + \hat{\mathbf{z}}E_z, \tag{3.111}$$

E 的拉普拉斯算子是

$$\nabla^2\mathbf{E} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\mathbf{E} = \hat{\mathbf{x}}\nabla^2 E_x + \hat{\mathbf{y}}\nabla^2 E_y + \hat{\mathbf{z}}\nabla^2 E_z. \tag{3.112}$$

因此，在笛卡尔坐标中，向量的拉普拉斯算子是其分量等于向量分量的拉普拉斯算子的向量。通过直接替换，可以

shown that

$$\nabla^2\mathbf{E} = \nabla(\nabla\cdot\mathbf{E}) - \nabla\times(\nabla\times\mathbf{E}). \tag{3.113}$$

Concept Question 3-11: What do the magnitude and direction of the gradient of a scalar quantity represent?

Concept Question 3-12: Prove the validity of Eq. (3.84c) in Cartesian coordinates.

Concept Question 3-13: What is the physical meaning of the divergence of a vector field?

Concept Question 3-14: If a vector field is solenoidal at a given point in space, does it necessarily follow that the vector field is zero at that point? Explain.

Concept Question 3-15: What is the meaning of the transformation provided by the divergence theorem?

Concept Question 3-16: How is the curl of a vector field at a point related to the circulation of the vector field?

Concept Question 3-17: What is the meaning of the transformation provided by Stokes's theorem?

Concept Question 3-18: When is a vector field "conservative"?

Chapter 3 Summary

Concepts

- Vector algebra governs the laws of addition, subtraction, and multiplication of vectors, and vector calculus encompasses the laws of differentiation and integration of vectors.
- In a right-handed orthogonal coordinate system, the three base vectors are mutually perpendicular to each other at any point in space, and the cyclic relations governing the cross products of the base vectors obey the right-hand rule.
- The dot product of two vectors produces a scalar, whereas the cross product of two vectors produces another vector.
- A vector expressed in a given coordinate system can be expressed in another coordinate system through the use of transformation relations linking the two coordinate systems.
- The fundamental differential functions in vector calculus are the gradient, the divergence, and the curl.
- The gradient of a scalar function is a vector whose magnitude is equal to the maximum rate of increasing change of the scalar function per unit distance, and its direction is along the direction of maximum increase.
- The divergence of a vector field is a measure of the net outward flux per unit volume through a closed surface surrounding the unit volume.
- The divergence theorem transforms the volume integral of the divergence of a vector field into a surface integral of the field's flux through a closed surface surrounding the volume.
- The curl of a vector field is a measure of the circulation of the vector field per unit area Δs, with the orientation of Δs chosen such that the circulation is maximum.
- Stokes's theorem transforms the surface integral of the curl of a vector field into a line integral of the field over a contour that bounds the surface.
- The Laplacian of a scalar function is defined as the divergence of the gradient of that function.

3-7 LAPLACIAN OPERATOR 163

表明

$$\nabla^2\mathbf{E} = \nabla(\nabla\cdot\mathbf{E}) - \nabla\times(\nabla\times\mathbf{E}). \tag{3.113}$$

概念问题3-11：标量梯度的大小和方向代表什么？概念问题 3-12：证明方程的有效性。 (3.84c) 在笛卡尔坐标中。概念问题3-13：矢量场散度的物理意义是什么？

Concept Question 3-14: If a vector field is solenoidal at a given point in space, does it necessarily follow that the vector field is zero at that point? Explain.

Concept Question 3-15: What is the meaning of the transformation provided by the divergence theorem?

Concept Question 3-16: How is the curl of a vector field at a point related to the circulation of the vector field?

Concept Question 3-17: What is the meaning of the transformation provided by Stokes's theorem?

Concept Question 3-18: When is a vector field "conservative"?

第三章小结

概念

- 向量代数控制向量的加法、减法和乘法定律，向量微积分包含向量的微分和积分定律。
- 在右手正交坐标系中，三个基向量在空间中的任意点都相互垂直，并且控制基向量叉积的循环关系遵循右手定则。
- 两个向量的点积产生一个标量，而两个向量的叉积产生另一个向量。
- 通过使用连接两个坐标系的变换关系，可以在给定坐标系中表达的向量在另一个坐标系中表达。
- 矢量微积分中的基本微分函数是梯度、散度和旋度。
- 标量函数的梯度是一个向量，其大小等于最大增加率标量函数每单位距离的变化，其方向是沿着增量最大的方向。
- 矢量场的散度是通过单位体积周围的封闭表面的每单位体积的净向外通量的度量。
- 散度定理将矢量场散度的体积积分转换为通过围绕体积的闭合表面的场通量的表面积分。
- 矢量场的旋度是每单位面积 Δs 矢量场循环的度量，选择 Δs 的方向以使循环最大。
- 斯托克斯定理将矢量场旋度的表面积分转换为该场在限定该表面的轮廓上的线积分。
- 标量函数的拉普拉斯算子定义为该函数梯度的散度。

Mathematical and Physical Models

Distance Between Two Points

$$d = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}$$

$$d = \left[r_2^2 + r_1^2 - 2r_1 r_2 \cos(\phi_2 - \phi_1) + (z_2 - z_1)^2\right]^{1/2}$$

$$d = \{R_2^2 + R_1^2 - 2R_1R_2[\cos\theta_2\cos\theta_1 + \sin\theta_1\sin\theta_2\cos(\phi_2 - \phi_1)]\}^{1/2}$$

Coordinate Systems Table 3-1

Coordinate Transformations Table 3-2

Vector Products

$$\mathbf{A}\cdot\mathbf{B} = AB\cos\theta_{AB}$$

$$\mathbf{A}\times\mathbf{B} = \hat{\mathbf{n}}\,AB\sin\theta_{AB}$$

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B})$$

$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C}) = \mathbf{B}(\mathbf{A}\cdot\mathbf{C}) - \mathbf{C}(\mathbf{A}\cdot\mathbf{B})$$

Vector Operators

$$\nabla T = \hat{\mathbf{x}}\frac{\partial T}{\partial x} + \hat{\mathbf{y}}\frac{\partial T}{\partial y} + \hat{\mathbf{z}}\frac{\partial T}{\partial z}$$

$$\nabla\cdot\mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$\nabla\times\mathbf{B} = \hat{\mathbf{x}}\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

(see Appendix C for cylindrical and spherical coordinates)

Divergence Theorem

$$\int_v \nabla\cdot\mathbf{E}\,dv = \oint_S \mathbf{E}\cdot d\mathbf{s}$$

Stokes's Theorem

$$\int_S (\nabla\times\mathbf{B})\cdot d\mathbf{s} = \oint_C \mathbf{B}\cdot d\mathbf{l}$$

Important Terms

Provide definitions or explain the meaning of the following terms:

azimuth angle
base vectors
Cartesian coordinate system
circulation of a vector
conservative field
cross product
curl operator
cylindrical coordinate system
differential area vector
differential length vector
differential volume
directional derivative
distance vector
divergence operator
divergence theorem
divergenceless
dot product
field lines
flux density
flux lines
gradient operator
irrotational field
Laplacian operator
magnitude
orthogonal coordinate system
position vector
radial distance r
range R
right-hand rule
scalar product
scalar quantity
simple product
solenoidal field
spherical coordinate system
Stokes's theorem
vector product
vector quantity
unit vector
zenith angle

PROBLEMS

Section 3-1: Vector Algebra

*3.1 Vector $\mathbf{A}$ starts at point $(1,-1,-3)$ and ends at point $(2,-1,0)$. Find a unit vector in the direction of $\mathbf{A}$.

*Answer(s) available in Appendix E.

3.2 Given vectors $\mathbf{A} = \hat{\mathbf{x}}2 - \hat{\mathbf{y}}3 + \hat{\mathbf{z}}$, $\mathbf{B} = \hat{\mathbf{x}}2 - \hat{\mathbf{y}} + \hat{\mathbf{z}}3$, and $\mathbf{C} = \hat{\mathbf{x}}4 + \hat{\mathbf{y}}2 - \hat{\mathbf{z}}2$, show that $\mathbf{C}$ is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.

*3.3 In Cartesian coordinates, the three corners of a triangle are $P_1 = (0,4,4)$, $P_2 = (4,-4,4)$, and $P_3 = (2,2,-4)$. Find the area of the triangle.

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数学与物理模型

两点间距离

$$d = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}$$

$$d = \left[r_2^2 + r_1^2 - 2r_1 r_2 \cos(\phi_2 - \phi_1) + (z_2 - z_1)^2\right]^{1/2}$$

$$d = \{R_2^2 + R_1^2 - 2R_1R_2[\cos\theta_2\cos\theta_1 + \sin\theta_1\sin\theta_2\cos(\phi_2 - \phi_1)]\}^{1/2}$$

坐标系 表3-1

坐标变换 表3-2

矢量积运算

$$\mathbf{A}\cdot\mathbf{B} = AB\cos\theta_{AB}$$

$$\mathbf{A}\times\mathbf{B} = \hat{\mathbf{n}}\,AB\sin\theta_{AB}$$

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B})$$

$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C}) = \mathbf{B}(\mathbf{A}\cdot\mathbf{C}) - \mathbf{C}(\mathbf{A}\cdot\mathbf{B})$$

矢量微分算子

$$\nabla T = \hat{\mathbf{x}}\frac{\partial T}{\partial x} + \hat{\mathbf{y}}\frac{\partial T}{\partial y} + \hat{\mathbf{z}}\frac{\partial T}{\partial z}$$

$$\nabla\cdot\mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$\nabla\times\mathbf{B} = \hat{\mathbf{x}}\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

圆柱坐标与球坐标下的表达式见附录 C。

散度定理

$$\int_v \nabla\cdot\mathbf{E}\,dv = \oint_S \mathbf{E}\cdot d\mathbf{s}$$

斯托克斯定理

$$\int_S (\nabla\times\mathbf{B})\cdot d\mathbf{s} = \oint_C \mathbf{B}\cdot d\mathbf{l}$$

重要术语

请给出下列术语的定义或解释其含义。

方位角
基矢量
直角坐标系
矢量的环量
保守场
叉积
旋度算子
圆柱坐标系
有向面元
有向线元
体积元
方向导数
两点间的位移矢量
散度算子
散度定理
无散性
点积
场线
通量密度
通量线
梯度算子
无旋场
拉普拉斯算子
模（大小）
正交坐标系
位置矢量
径向距离 r
球径 R
右手定则
标量积
标量
标量与矢量的乘积
管形场（无散场）
球坐标系
斯托克斯定理
矢量积
矢量
单位矢量
天顶角

问题

第 3-1 节：向量代数

*3.1 向量 A 从点 (1, −1, −3) 开始，到点 (2, −1, 0) 结束。求 A 方向的单位向量。

* 标记的习题在附录 E 提供答案。

3.2 给定向量 A = x̂2 − ŷ3 + ẑ、B = x̂2 − ŷ + ẑ3 和 C = x̂4 + ŷ2 − ẑ2，表明 C 垂直于 A 和 B。

*3.3 在笛卡尔坐标系中，三角形的三个角为 $P_1 = (0, 4, 4)$，$P_2 = (4, -4, 4)$ 和 $P_3 = (2, 2, -4)$。求三角形的面积。

3.4 Given $\mathbf{A} = \hat{\mathbf{x}}2 - \hat{\mathbf{y}}3 + \hat{\mathbf{z}}1$ and $\mathbf{B} = \hat{\mathbf{x}}B_x + \hat{\mathbf{y}}2 + \hat{\mathbf{z}}B_z$:

(a) Find B_x and B_z if $\mathbf{A}$ is parallel to $\mathbf{B}$.

(b) Find a relation between B_x and B_z if $\mathbf{A}$ is perpendicular to $\mathbf{B}$.

3.5 Given vectors $\mathbf{A} = \hat{\mathbf{x}} + \hat{\mathbf{y}}2 - \hat{\mathbf{z}}3$, $\mathbf{B} = \hat{\mathbf{x}}2 - \hat{\mathbf{y}}4$, and $\mathbf{C} = \hat{\mathbf{y}}2 - \hat{\mathbf{z}}4$, find the following:

*(a) A and $\hat{\mathbf{a}}$

(b) The component of $\mathbf{B}$ along $\mathbf{C}$

(c) θ_{AC}

(d) $\mathbf{A} \times \mathbf{C}$

*(e) $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$

(f) $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$

(g) $\hat{\mathbf{x}} \times \mathbf{B}$

*(h) $(\mathbf{A} \times \hat{\mathbf{y}}) \cdot \hat{\mathbf{z}}$

3.6 Given vectors $\mathbf{A} = \hat{\mathbf{x}}2 - \hat{\mathbf{y}} + \hat{\mathbf{z}}3$ and $\mathbf{B} = \hat{\mathbf{x}}3 - \hat{\mathbf{z}}2$, find a vector $\mathbf{C}$ whose magnitude is 9 and whose direction is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.

3.7 Given $\mathbf{A} = \hat{\mathbf{x}}(x+2y) - \hat{\mathbf{y}}(y+3z) + \hat{\mathbf{z}}(3x-y)$, determine a unit vector parallel to $\mathbf{A}$ at point $P = (1,-1,2)$.

3.8 By expansion in Cartesian coordinates, prove:

(a) The relation for the scalar triple product given by Eq. (3.29).

(b) The relation for the vector triple product given by Eq. (3.33).

*3.9 Find an expression for the unit vector directed toward the origin from an arbitrary point on the line described by $x = 1$ and $z = -2$.

3.10 Find an expression for the unit vector directed toward the point P located on the z axis at a height h above the x–y plane from an arbitrary point $Q = (x, y, -5)$ in the plane $z = -5$.

*3.11 Find a unit vector parallel to either direction of the line described by

$$2x + z = 4$$

3.12 Two lines in the x–y plane are described by the following expressions:

$$\begin{array}{ll} \text{Line 1} & x + 2y = -6 \\ \text{Line 2} & 3x + 4y = 8 \end{array}$$

Use vector algebra to find the smaller angle between the lines at their intersection point.

*3.13 A given line is described by

$$x + 2y = 4.$$

Vector $\mathbf{A}$ starts at the origin and ends at point P on the line such that $\mathbf{A}$ is orthogonal to the line. Find an expression for $\mathbf{A}$.

3.14 Show the following given two vectors $\mathbf{A}$ and $\mathbf{B}$:

(a) The vector $\mathbf{C}$ defined as the vector component of $\mathbf{B}$ in the direction of $\mathbf{A}$ is given by

$$\mathbf{C} = \hat{\mathbf{a}}(\mathbf{B} \cdot \hat{\mathbf{a}}) = \frac{\mathbf{A}(\mathbf{B} \cdot \mathbf{A})}{|\mathbf{A}|^2},$$

where $\hat{\mathbf{a}}$ is the unit vector of $\mathbf{A}$.

(b) The vector $\mathbf{D}$ defined as the vector component of $\mathbf{B}$ perpendicular to $\mathbf{A}$ is given by

$$\mathbf{D} = \mathbf{B} - \frac{\mathbf{A}(\mathbf{B} \cdot \mathbf{A})}{|\mathbf{A}|^2}$$

*3.15 A certain plane is described by

$$2x + 3y + 4z = 16$$

Find the unit vector normal to the surface in the direction away from the origin.

3.16 Given $\mathbf{B} = \hat{\mathbf{x}}(z-3y) + \hat{\mathbf{y}}(2x-3z) - \hat{\mathbf{z}}(x+y)$, find a unit vector parallel to $\mathbf{B}$ at point $P = (1,0,-1)$.

*3.17 Find a vector $\mathbf{G}$ whose magnitude is 4 and whose direction is perpendicular to both vectors $\mathbf{E}$ and $\mathbf{F}$, where $\mathbf{E} = \hat{\mathbf{x}} + \hat{\mathbf{y}}2 - \hat{\mathbf{z}}2$ and $\mathbf{F} = \hat{\mathbf{y}}3 - \hat{\mathbf{z}}6$.

3.18 A given line is described by the equation:

$$y = x - 1.$$

Vector $\mathbf{A}$ starts at point $P_1 = (0,2)$ and ends at point P_2 on the line, at which $\mathbf{A}$ is orthogonal to the line. Find an expression for $\mathbf{A}$.

3.19 Vector field $\mathbf{E}$ is given by

$$\mathbf{E} = \hat{\mathbf{R}}\,5R\cos\theta - \hat{\boldsymbol{\theta}}\frac{12}{R}\sin\theta\cos\phi + \hat{\boldsymbol{\phi}}3\sin\phi.$$

Determine the component of $\mathbf{E}$ tangential to the spherical surface $R = 2$ at point $P = (2, 30°, 60°)$.

3.20 When sketching or demonstrating the spatial variation of a vector field, we often use arrows, as in **Fig. P3.20**, wherein the length of the arrow is made to be proportional to the strength of the field and the direction of the arrow is the same as that of the field's. The sketch shown in **Fig. P3.20**, which represents the vector field $\mathbf{E} = \hat{\mathbf{r}}r$, consists of arrows pointing radially away from the origin and their lengths increasing linearly in proportion to their distance away from the origin.

3.4 给定 A = x̂2 − ŷ3 + ẑ1 和 B = x̂Bx + ŷ2 + ẑBz：
(a) 如果 A 与 B 平行，求 Bx 和 Bz。
(b) 如果 A 垂直于 B，求 Bx 和 Bz 之间的关系。

3.5 给定向量 A = x̂ + ŷ2 − ẑ3、B = x̂2 − ŷ4 和 C = ŷ2 − ẑ4，求出以下结果：
*(a) A 和 â (b) B 沿 C 的分量 (c) θAC d) A× C *(e) A ·(B× C)f) A (× (B× C)g) x̂× B *(h) (A× ŷ ( )·ẑ

3.6 给定向量 A = x̂2 − ŷ + ẑ3 和 B = x̂3 − ẑ2，找到一个向量 C，其大小为 9，方向垂直于 A 和 B。

3.7 给定 A = x̂(x + 2y) − ŷ(y + 3z) + ẑ(3x − y)，确定在点 P = (1, −1, 2) 处与 A 平行的单位向量。

3.8 通过笛卡尔坐标展开，证明：
(a) 等式给出的标量三重积的关系。 (3.29) 。 (b) 由方程给出的向量三重积的关系。 (3.33) 。

*3.9 找出从 x = 1 和 z = −2 描述的直线上的任意点指向原点的单位向量的表达式。

3.10 找到从平面 z = −5 中的任意点 Q = (x, y, −5) 指向位于 x-y 平面上方高度 h 的 z 轴上的点 P 的单位向量的表达式。

*3.11 求一个与下式描述的线的任一方向平行的单位向量

$$2x + z = 4$$

3.12 x-y 平面中的两条直线由以下表达式描述：

$$\begin{array}{ll} \text{Line 1} & x + 2y = -6 \\ \text{Line 2} & 3x + 4y = 8 \end{array}$$

使用向量代数找到线交点处的较小角度。

*3.13 给定的线由下式描述

$$x + 2y = 4.$$

矢量 A 从原点开始，到线上的点 P 结束，使得 A 与该线正交。找出 A 的表达式。

3.14 显示以下给定的两个向量 A 和 B：
(a) 向量 C 定义为 B 在 A 方向上的向量分量，由下式给出

$$\mathbf{C} = \hat{\mathbf{a}}(\mathbf{B} \cdot \hat{\mathbf{a}}) = \frac{\mathbf{A}(\mathbf{B} \cdot \mathbf{A})}{|\mathbf{A}|^2},$$

其中 â 是 A 的单位向量。
(b) 向量 D 定义为垂直于 A 的 B 的向量分量，由下式给出

$$\mathbf{D} = \mathbf{B} - \frac{\mathbf{A}(\mathbf{B} \cdot \mathbf{A})}{|\mathbf{A}|^2}$$

*3.15 某个平面被描述为

$$2x + 3y + 4z = 16$$

求远离原点方向上垂直于表面的单位向量。

3.16 给定 B = x̂(z − 3y) + ŷ(2x − 3z) − ẑ(x + y)，求在点 P = (1, 0, −1) 处平行于 B 的单位向量。

*3.17 求一个向量G，其大小为4，方向垂直于向量E和F，其中E = x̂ + ŷ2 − ẑ2和F = ŷ 3 − ẑ6。

3.18 给定的直线由以下方程描述：

$$y = x - 1.$$

矢量 A 从线上的点 P1 = (0, 2) 开始，到点 P2 结束，在该点 A 与线正交。找出 A 的表达式。

3.19向量场E由下式给出

$$\mathbf{E} = \hat{\mathbf{R}}\,5R\cos\theta - \hat{\boldsymbol{\theta}}\frac{12}{R}\sin\theta\cos\phi + \hat{\boldsymbol{\phi}}3\sin\phi.$$

确定 E 在点 P = (2、30°、60°) 处与球面 R = 2 相切的分量。

3.20 当描绘或演示矢量场的空间变化时，我们经常使用箭头，如图 P3.20 所示，其中箭头的长度与场的强度成正比，并且箭头的方向与场的方向相同。图 P3.20 所示的草图表示矢量场 E = r̂r，由远离原点的径向箭头组成，它们的长度与远离原点的距离成正比线性增加。